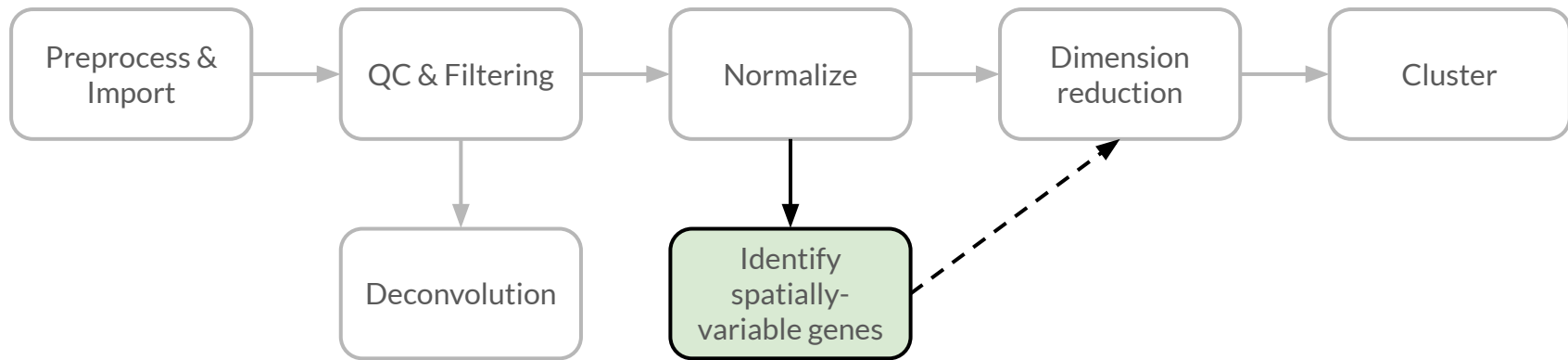




Identifying spatially-variable genes

Data Lab

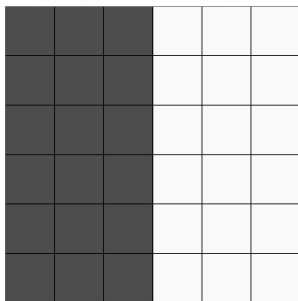


- Highly variable genes are good, but are they always what we really want to know about?
- If tissues are structured (they usually are) then we would expect structured variation in gene expression
- Identifying spatially variable genes will allow us to better understand the biologically relevant changes in gene expression across the tissue
- Can be used in place of HVGs as the first step in dimension reduction/clustering

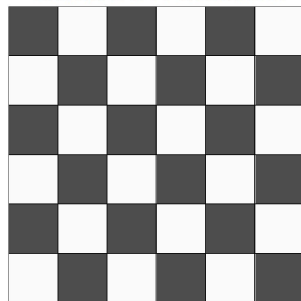
Quantifying spatial variation

- To quantify spatial variation, we combine gene expression data with spatial information, usually calculating some version of *spatial autocorrelation*
 - Many possible measures, with a lot of classical theory from work in geography

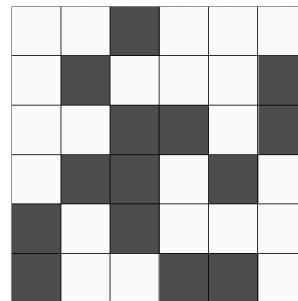
Positive spatial autocorrelation



Negative spatial autocorrelation



No spatial autocorrelation

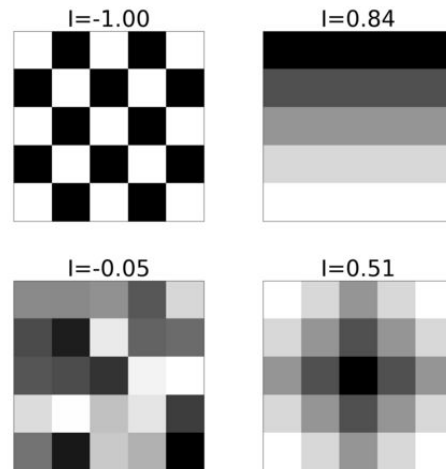


Moran's I

- One of the oldest methods of quantifying spatial variation is Moran's I ([Moran 1950](#))

$$I = \frac{n \sum_{ij} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_{ij} w_{ij} \times \sum_i (x_i - \bar{x})^2}$$

- The weights are a function of distance
 - closer elements get higher weights
 - often, only direct neighbors are considered, everything else gets weight 0
- Moran's I values range from -1 to 1
 - We will rarely see negative values



https://en.wikipedia.org/wiki/Moran%27s_I

Methods for spatial gene expression

- *Many* methods have been developed to apply spatial statistics to gene expression data
 - Goals: account for particular processes and expected distributions in spatial data: Sparse data, counts, etc.
 - Differences in statistical methods and goals
 - categorized by [Yan, Hua, & Li \(2025\)](#) (see table)
- With so many options, how to choose?
 - Computational efficiency, accuracy, ease of use
 - [Li et al. \(2026\)](#) reviewed many
 - SPARK-X, SpatialDE & nnSVG performed best
 - Moran's I not far behind

Table 1 | Classification of 34 SVG detection methods along with statistical inference types

Inference types	Categories	Overall SVGs	Cell-type-specific SVGs	Spatial-domain-marker SVGs
Frequentist inference	Dependence tests	SPARK-X		
		SMASH		
		singlecellHaystack		
		Trendsceek		
		HEARTSVG		
		Hotspot		
		MERINGUE		
		SpaGene		
		BinSpect		
		scGCO		
	Regression fixed-effect tests	RayleighSelection		
		SPADE	CTSV C-SIDE spVC	SpaGCN DESpace
	Regression random-effect tests	SpatialDE		
		nnSVG		
		SOMDE		
		SVCA		
		SPARK		
		GPcounts		
Bayesian inference		BOOST-GP		
		BOOST-MI		
		BOOST-HMI		
No statistical inference		MULTILAYER		
		sepal		
		BSP ^a		
		PROST ^b		
		HRG		
		SVGbit		
		SINFONIA		
		SpaGFT ^c		

nnSVG

- We will be using nnSVG ([Weber et al. 2023](#)) for an alternative analysis to Moran's I
 - Fairly high accuracy in comparison to other methods
 - Fairly computationally efficient (SPARK-X is better)
 - Part of Bioconductor (maintained!)
 - Is it the “best” method? Probably not, especially as new methods arise
- How does nnSVG work?
 - Identify nearest neighbors for each spot (or cell)
 - Fit a model of spatial variation (nearest-neighbor Gaussian process, for the technically-minded)
 - Model includes a length parameter that allows for different scales of covariation for different genes
 - Using a fixed number of cells improves efficiency

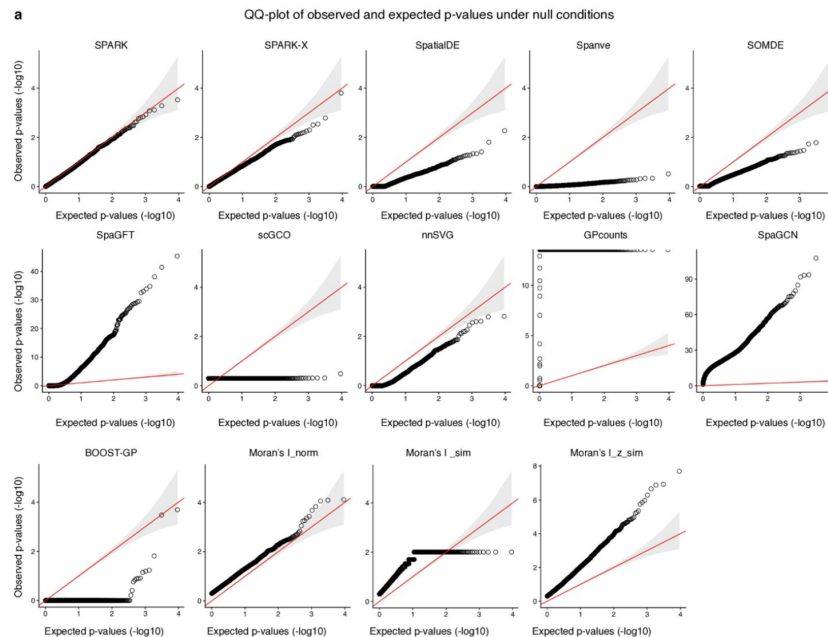
nnSVG statistics

The output of **nnSVG** is a set of statistics for each gene including:

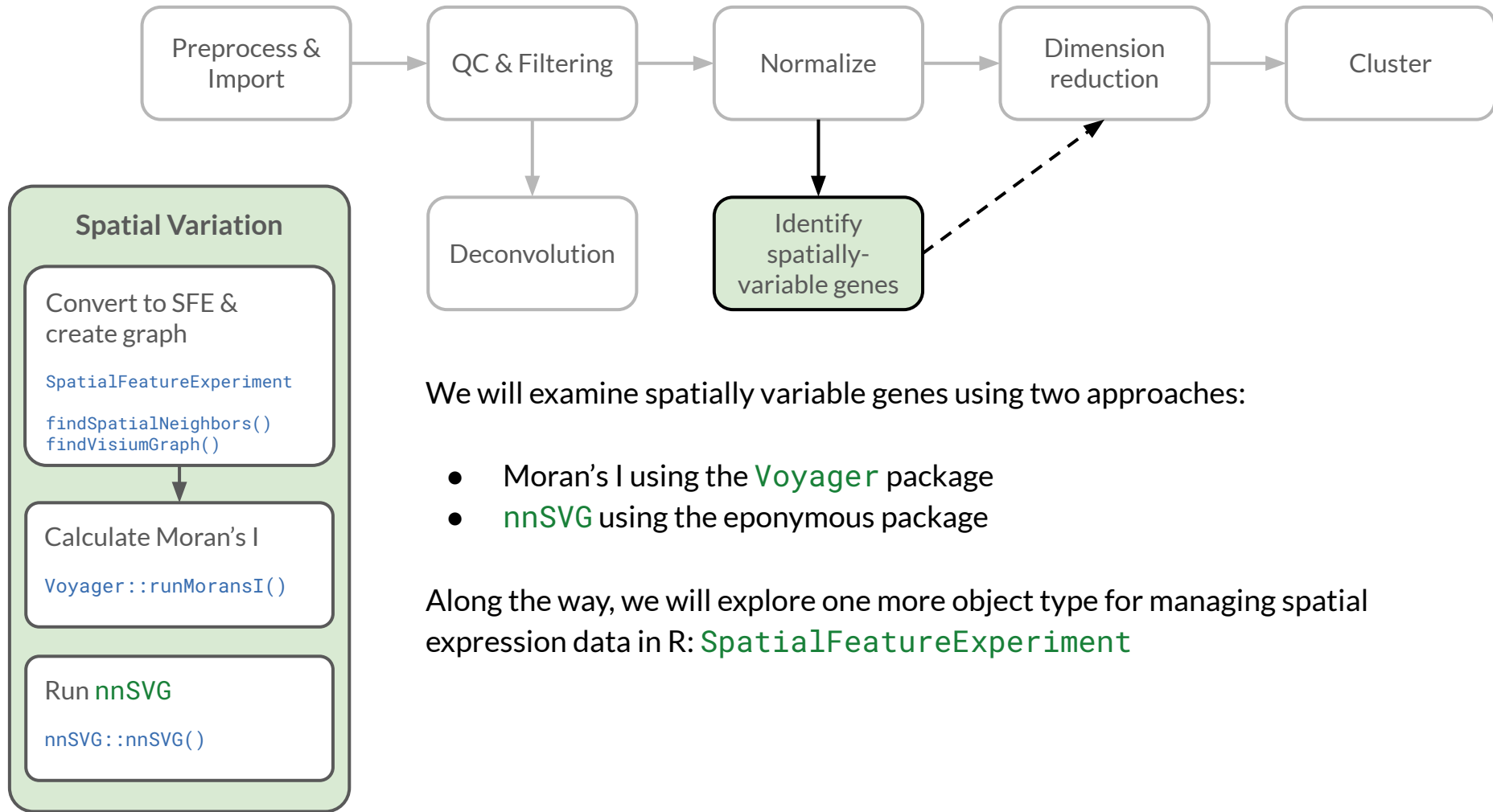
- the estimated length scale parameter (**phi**)
- a likelihood ratio comparing the spatial model to a non-spatial model (**LR_stat**)
- an estimate of the proportion of spatial variation (**prop_sv**)
- P value (**pval**) and adjusted P value (**padj**)
- a variety of intermediate statistics that go toward those estimates
 - **sigma.sq**: estimate of spatial variation
 - **tau.sq**: estimate of non-spatial variation

Be careful!

- P values are hard to get right; many methods overestimate them
 - Often better to just use ranks rather than trusting P values
 - Beware rotation!
 - Methods *should not* give different results depending on orientation of the slide
 - Sadly, some do (including SPARK-X under some recommended settings)
- See [Su & Cui \(2025\)](#)



from [Li et al. \(2026\)](#)



SpatialFeatureExperiment objects

- Most of `SpatialFeatureExperiment` (SFE) is the same as the `SpatialExperiment` class we have been using up to this point
- What SFE adds is a representation of the *graph* that connects spots or cells
 - Graph can be created using preset relationships for Visium data
 - Or use a nearest neighbor search to find connections for each spot/cell
 - This graph allows efficient, rotation-invariant, calculations
- More documentation at the package website:
<https://pachterlab.github.io/SpatialFeatureExperiment/>

